Exposure Fusion-Based Shadow-Insensitive Hyperspectral Target Detection

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***Abstract*— Hyperspectral images (HSIs) have been widely used for target detection due to their abundant spatial and spectral information. In this article, a shadow-insensitive hyperspectral target detection (HTD) framework based on exposure fusion is proposed, which consists of the following major steps. First, the input HSI is divided into two parts, namely the shadow region and the nonshadow region. Second, total variation-based feature extraction and overexposure operation are performed on the input image to produce two feature images, i.e., the original feature image and the overexposure image. Third, a self-guided constrained energy minimization (SGCEM) detector is performed on the two feature images to detect the targets in shadow and nonshadow regions, respectively. Finally, the detection results obtained on the original feature image and the overexposure image are fused to acquire the final detection result. Extensive experiments conducted on real-world data illustrate that the proposed method can achieve satisfactory results when shadow exists.**

***Index Terms*— Exposure fusion, hyperspectral image (HSI), shadow, target detection.**

I. Introduction

RICH spatial and surface spectral information are recorded in hyperspectral images (HSIs), which has aroused researchers' interest in HSI processing, such as target detection, anomaly detection, land cover classification, spectral unmixing, and so on [1], [2], [3], [4], [5]. In these research topics, hyperspectral target detection (HTD) aims to separate the target pixels from the background based on their inherent spectral differences, which usually uses the spectrum of the target as the prior information. HTD can be used to detect camouflaged targets or targets hidden in complex backgrounds [6], which has drawn great attention recently and performs an important function in practical applications, such as mineral exploration, military reconnaissance, and environmental monitoring [7], [8], [9], [10]. Although many HTD methods have been proposed, HTD still faces the significant challenge of spectra variability caused by many issues, such as the presence of shadows and fog [11], [12].

In the early stage, scholars calculate the spectral similarity between the target and tested pixels to realize detection. The classic spectral similarity measurement methods include spectral angle mapping (SAM) [13], Euclidean distance (ED) [14], and spectral information divergence (SID) [15]. These methods are simple but do not take the spatial relationship of pixels into account. In order to utilize the background information in HSIs, constrained statistical assumptions (e.g., linear mixing model and Gaussian distribution) are introduced in HTD. Among these statistical-based methods, adaptive cosine/coherent estimator (ACE) [16] and match filter (MF) [17] are the most representative methods. ACE and MF are derived from a generalized likelihood ratio test and binary hypothesis testing. A series of improved methods are also proposed, such as regularized spectral matched filter (RMF) [18] and multitarget multiple-instance adaptive cosine estimator (MTMI-MF) [19].

Furthermore, constrained energy minimization (CEM) [20] is a classic detector that constructs a filter to minimize the output energy while highlighting targets. Many extended versions based on CEM are widely developed to suppress background. For instance, hierarchical CEM (hCEM) utilizes a layer-by-layer filtering procedure for background suppressing [21]. Ensemble-based CEM (ECEM) [22] is proposed to improve the detection nonlinearity and generalization ability of CEM. Chen et al. [6] design a global to local hierarchical detection algorithm (G2LHTD) with a diverse-direction CEM (D^2CEM) detector which can make more use of local spatial information within eight neighborhoods around each test pixel.

Considering that the above methods rely on statistical assumptions or do not fully utilize spatial information in HSIs, machine learning (ML)-based detection methods are investigated. As one of the most typical ML methods, sparse representation detectors use the sparsity of target and construct a dictionary for detection. For example, combined sparse and collaborative representation (CSCR) is developed for HTD in [23]. Chen et al. [24] realize the sparse representation

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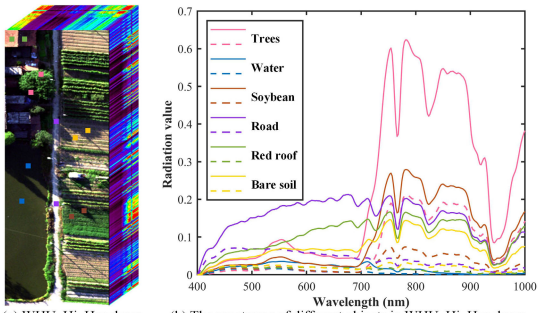
by solving a constrained optimization problem. Then, the characteristics of the sparse vector are utilized to obtain the detection result. Although this kind of methods has attained satisfactory results, its performance highly relies on sufficient prior target information to meet the requirements of dictionary construction [25].

In a data-driven manner, deep learning (DL)-based HTD algorithms constantly sprang up [25], [26], [27], [28], [29]. The current DL-based detection algorithms mainly focus on how to solve the training sample size problem, since the prior target samples are scarce and cannot meet the requirement of training usually. For instance, a modified autoencoder (AE) and linear prediction strategy are developed in [26] to augment the training samples. Compared with target samples, background samples are easily obtained and have a much larger size. Thus, Xie et al. [27] train the model with the background spectral samples to learn the background distribution. After that, the discrepancy between the reconstructed and original HSIs can be used to find the targets. Moreover, some DL-based methods try to train the model in an unsupervised way and use a matching strategy to spot the target [30], [31]. In general, DL-based methods can provide better performance than traditional methods when the training data are sufficient. But DL-based methods typically suffer from high computational complexity, which is not suitable for in-orbit processing [6].

Apart from the above methods, saliency detection is often used for object detection in remote sensing images, which aims to locate the most prominent objects [32], [33], [34], [35]. For example, Huang et al. [32] utilize a saliency mechanism to construct a discriminant dictionary for detection. Ren et al. [33] combine superpixel representation and PageRank algorithm to generate a saliency map for object detection. Although saliency detection is successfully applied in object detection, most saliency methods do not take illumination variation into account. These methods may not perform well when the intensity of targets is weakened with lower illumination.

On the whole, the existing target detection methods are fruitful. However, spectral variation in real-world scenes is still a serious challenge in HTD with deep research [36]. Generally speaking, the prior knowledge required for HTD is one target spectrum, which is very limited [25]. Thus, most of the developed HTD techniques are sensitive to the prior target spectrum due to spectral variation. The spectral variation problem is inevitable, especially when illumination changes. For instance, Fig. 1 presents the spectrums of different objects in the WHU_Hi_Hanchuan data [37]. Obviously, there are significant differences in the spectrums of the same object that exists in shadow and nonshadow regions. In addition, shadows not only reduce the radiation value of ground objects, but also increase the difficulty of distinguishing different objects. It is evident that the impact of illumination variation on spectrums cannot be ignored.

Due to the occlusion of light by objects, such as buildings, trees, and clouds, shadows exist in remote sensing images inevitably. Thus, an effective target detection algorithm should be robust to illumination variation. To solve the spectral variation problem caused by shadows, some physics-based methods are proposed. In [38], the spectrum in the



(a) WHU_Hi_Hanchuan (b) The spectrums of different objects in WHU_Hi_Hanchuan

Fig. 1. Spectrums of different objects under well-illumination conditions (solid line) and the shadows (dashed line). The color blocks in (a) denote the pixels presented in (b). (a) WHU_Hi_Hanchuan. (b) Spectrums of different objects in WHU_Hi_Hanchuan.

nonshadow region and the shadowed spectrums generated by a physics-based model are sent to a spectral angle-stacked AE for shadow-insensitive feature learning. Zhang et al. [3] analyze the influence of shadow on reflectance and develop a physics-based spectral mixture model for shadow-aware spectral unmixing. Moreover, the radiation transport model is utilized to generate the shadowed spectrum. In [39], moderate transmission (MODTRAN) model is used to compute a skylit-to-fully illuminated spectral ratio for transforming a sunlit spectrum of the chosen material to a spectrum in shadows accordingly. The original and the shadowed spectrums are incorporated for detection or classification subsequently. Wiseman et al. [40] also utilize MODTRAN and fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) models to simulate the spectrums under different illumination conditions for ACE detector. Similarly, the spectrum under well-illuminated conditions is converted into a shadowed spectrum with "whitening/dewhitening" of target signatures and covariance matrices in [41]. The physics-based methods can generate reliable shadowed spectrum for detection, but physics model parameters are difficult to estimate accurately, which may make this kind of methods inapplicable in real-world scenes.

Apart from the aforementioned researches, shadow interference in other HSI processing tasks also has been studied. For instance, Cui and Prasad [42] attempt to find a lower dimensional subspace which is robust to the change of illumination for HSI classification. Yang [43] believes that sunlit and shadowed components in each pixel are integrated by a bilinear mixing mechanism. Then, a supervised nonlinear unmixing method with automatic shadow compensation is explored. Besides, Wan et al. [44] prove that the phase correlation is robust to illumination variation, and design an illumination-invariant change detection metric based on dense phase correlation matching. Overall, only a few methods are proposed for shadow-insensitive HTD. The spectral variation caused by shadow is not well solved in HTD right now. Besides, the experimental data are limited. More data are needed to validate the effectiveness of the detection methods.

As a mature technology, multiexposure fusion is widely utilized to deal with the illumination variation [45], [46], which has been utilized to remove the shadow in HSIs [46]. It brings the idea that spectrum signals in shadows can be recovered

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by generating an overexposure HSI. Therefore, this work utilizes exposure fusion to solve the spectral variation problem caused by lower illumination. In the proposed method, the dimension reduction and feature extraction are operated on the HSI at first. Then, a self-guided CEM (SGCEM) detector is used to detect the targets in the nonshadow regions with the feature image and the target located in shadow regions with the overexposure feature image. Finally, the initial detection results are fused together to obtain the final detection result.

The major contribution of this work is providing a novel method for target detection under illumination variation conditions. Specifically, this work is a successful practice to relight the shadows with exposure technique rather than model the shadowed spectrum for detection. Furthermore, the proposed method is parameter-free and constructs a modified CEM detector which uses the preliminary detection result to improve CEM. Besides, eight HSIs both containing targets in shadow and nonshadow regions, are adopted to confirm the application potential of the proposed method.

II. CEM Detector

In this section, the well-known CEM detector is briefly reviewed. Motivated by the principle of CEM, an improved CEM (called SGCEM) is provided in our work, which will be introduced in Section III.

CEM is widely used in HSI target detection tasks. The principle of CEM is to construct a finite impulse filter (FIR) to minimize the total energy of the output. Meanwhile, the response of the target signature is constant. The HSI can be rearranged as a 2-D matrix $\mathbf{X} \in \mathbb{R}^{M\times D}$, $\mathbf{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_M]$, $\boldsymbol{x}_i$ represents the spectral vector of pixel i. M and D are the sum number of pixels in each band and the number of bands in the HSI, respectively. In fact, the linear filter of CEM is realized by a set of weights and calculated as follows:

$$\boldsymbol{y} = \boldsymbol{w}^T \boldsymbol{x} \tag{1}$$

where $\boldsymbol{y}$ is the detection output, which can be considered as the inner product of the input spectrum $\boldsymbol{x}$ and the weight vector $\boldsymbol{w}$. As introduced before, the weight vector $\boldsymbol{w}$ is estimated by minimizing the energy of output $\boldsymbol{y}$, which can be converted into solve the optimization problem as follows:

$$\begin{cases} \arg\min\limits_{\boldsymbol{w}} \dfrac{1}{M}\left(\sum\limits_{i=1}^{M} \boldsymbol{y}_i^2\right) = \arg\min\limits_{\boldsymbol{w}} \boldsymbol{w}^T \mathbf{R} \boldsymbol{w} \\ \text{s.t. } \boldsymbol{w}^T \boldsymbol{t} = 1 \end{cases} \tag{2}$$

where $\boldsymbol{t}$ is the known target spectrum. $\mathbf{R} \in \mathbb{R}^{D\times D}$ is the maximum likelihood estimation of the correlation matrix related to the HSI, which can be computed by all pixels in the image as $\mathbf{R} = (1/M)\sum \boldsymbol{x}_i \boldsymbol{x}_i^T$. Besides, the solution of the optimization problem is given by

$$\boldsymbol{w}' = \frac{\mathbf{R}^{-1}\boldsymbol{t}}{\boldsymbol{t}^T \mathbf{R}^{-1} \boldsymbol{t}}. \tag{3}$$

According to the above equations, it can be seen that the detection result is highly decided by the correlation matrix $\mathbf{R}$.

III. Proposed Method

This section provides an exhaustive introduction to the proposed method. Since the target spectrums are quite distinct in different illumination conditions, different detection strategies are adopted for different illumination regions. As displayed in Fig. 2, the proposed method mainly contains two procedures, namely target detection in shadow and nonshadow regions and decision fusion. Besides, the detection process is finished by SGCEM.

A. Target Detection in Shadow and Nonshadow Regions

This step aims to obtain the detection results of different illumination regions, which can be further divided into three parts, namely shadow detection, target detection in the nonshadow region, and target detection in the shadow region.

1) Shadow Detection: Shadow detection results can provide the basis for subsequent detection process. To get the shadow mask, a multiple threshold method is proposed here. Specifically, four shadow feature images which have relatively lower reflectivity in the shadow region are extracted. Then, binary processing is executed on these shadow feature images with adaptive thresholds to obtain shadow map candidates. Finally, these shadow maps are combined to get the shadow mask with maximum voting.

Denoting the HSI with a known wavelength range as $\mathbf{I}$, the red, green, blue, and near infrared band (NIR) images can be represented as $\mathbf{I}_R$, $\mathbf{I}_G$, $\mathbf{I}_B$, and $\mathbf{I}_{\text{NIR}}$. Since shadow regions have much lower pixel values in the RGB and NIR band [47], $\mathbf{I}_{\text{NIR}}$ is considered as the first shadow feature image $\mathbf{S}_1$. Then, the real RGB image is converted to hue-saturation-value (HSV) color space. The brightness image $\mathbf{B}$ corresponding to the value component is the second shadow feature image $\mathbf{S}_2$. The third image $\mathbf{S}_3$ is the color index $(\mathbf{I}_G - \mathbf{I}_B)/(\mathbf{I}_G + \mathbf{I}_B)$. The fourth image $\mathbf{S}_4$ is set as $\mathbf{I}_{\text{NIR}} + \mathbf{B} + \mathbf{I}_B$.

In the above shadow feature images, shadow pixels have much lower pixel values, which indicate that the shadow candidates can be separated through binarization. Actually, the automatic thresholding method Otsu [48] is used to acquire the binarized results. Moreover, each binarized result is optimized with the morphological opening operation which can remove some wrong shadow pixels. Assuming the binarized shadow candidates corresponding to above four shadow feature images are $\hat{\mathbf{S}}_1$, $\hat{\mathbf{S}}_2$, $\hat{\mathbf{S}}_3$, and $\hat{\mathbf{S}}_4$. Then, all shadow maps are summed to decide the final shadow mask as follows:

$$\begin{aligned} \mathbf{S} &= 2 \times \hat{\mathbf{S}}_1 + \hat{\mathbf{S}}_2 + \hat{\mathbf{S}}_3 + \hat{\mathbf{S}}_4 \\ \mathbf{M}_S &= \begin{cases} 1, & \text{if } \mathbf{S} \geq 3 \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \tag{4}$$

According to the shadow mask $\mathbf{M}_S$, the image can be divided into the nonshadow region $\mathbf{M}_N$ and the shadow region $\mathbf{M}_S$, nonshadow region also represents well-illuminated region. It is worth to mentioned that $\hat{\mathbf{S}}_1$, the shadow map of $\mathbf{I}_{\text{NIR}}$, is considered twice due to its importance in shadow detection.

Multiple threshold constraints on multiple images can reduce error rates in shadow detection. Furthermore, the purpose of shadow detection is to generate a rough shadow mask for generating the overexposure image and decision fusion,

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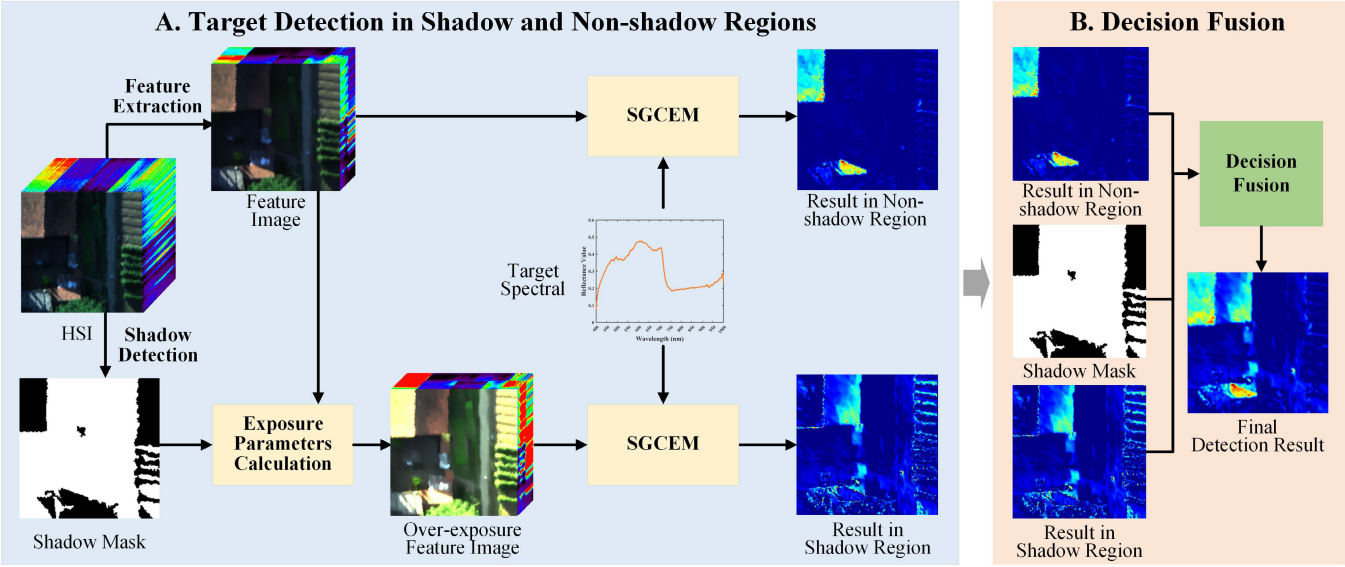


Fig. 2. Flowchart of the proposed framework. (a) Target detection in shadow and nonshadow regions. (b) Decision fusion.

rather than accurately detect shadows. It means an efficient shadow detection method that can get a broadly right shadow mask is sufficient for the proposed detection framework.

2) Target Detection in the Nonshadow Region: In the nonshadow region, the variety of spectrums caused by illumination is negligible, which means the spectrums of potential targets are consistent with the input spectrum relatively. Thus, the HSI can be directly used for detection in the nonshadow region as a regular detection process. However, the original HSI $\mathbf{I}$ with a size of $R \times C \times D$ is a high dimensional structure. $R \times C$ means the spatial size of the image, while D is the spectral dimension. To reduce the redundant information while maintaining the characteristics of original spectral curves, band averaging method is performed on $\mathbf{I}$ to get the dimension reduced image $\mathbf{I}_k^{DR}, k \in \{1, \ldots, K\}$ by (5). In this work, K is set as the up-rounded value of $D/10$

$$\mathbf{I}_k^{DR} = \begin{cases} (\mathbf{I}_{((k-1)\lfloor D/K \rfloor+1)}, \ldots, \mathbf{I}_{(k\lfloor D/K \rfloor)}), & \text{if } k\lfloor D/K \rfloor \le D \\ (\mathbf{I}_{(K-\lfloor D/K \rfloor+1)}, \ldots, \mathbf{I}_D), & \text{if } k\lfloor D/K \rfloor > D. \end{cases} \tag{5}$$

Then, feature extraction is executed on the dimension-reduced image $\mathbf{I}^{DR}$ to improve the spectral purity among pixels. Specifically, a relative total variation (RTV) technique [49] is used to extract the feature image $\mathbf{F}$ of $\mathbf{I}^{DR}$, since RTV is able to remove the noise and texture in the image [50].

The feature image $\mathbf{F}$ is then fed into the SGCEM to acquire the distribution map of target in the nonshadow region. The SGCEM can be considered as a double detection process that uses different correlation matrices to design the filter. At the first detection process, the correlation matrix $\mathbf{R}$ is calculated by all pixels in feature image $\mathbf{F}$. Then, the first detection result can be binarized with the Ostu method [48] to find some high-confidence target pixels. These target pixels are removed when computing the correlation matrix $\mathbf{R}$ to modify CEM detector at the second detection process. Finally, the modified CEM detector which can better suppress the background is performed on $\mathbf{F}$ to obtain the detection result $\mathbf{P}_N$.

3) Target Detection in the Shadow Region: In the shadow region, pixels have lower values due to insufficient illumination. To improve the brightness of HSI, an overexposure image simulation technique is utilized to relight the shadow areas. More exactly, the purpose of generating the overexposure image is to restore the spectral reflectance of pixels in shadows and make it similar to the spectral reflectance of pixels in the nonshadow region. Besides, exposure parameters are generated adaptively according to the brightness difference between pixels in nonshadow and shadow areas. The overexposure image is produced by pixelwise multiplication with the exposure parameters and original feature image.

The specific exposure parameters calculation process is introduced below. At first, the averaged reflectance value of each band $\boldsymbol{q}_j^s, j \in \{1, \ldots, D\}$ in shadows is calculated by

$$\boldsymbol{q}_j^s = \sum_{i=1}^{N_1} (\mathbf{I}_j \cdot \mathbf{M}_S) \Big/ N_1 \tag{6}$$

where j is the band number, N_1 is the sum number of pixels in shadows, and $\cdot$ means elements dot multiplication. Then, the average value of pixels around shadows in the nonshadow region is calculated. To find these pixels, dilation is operated on the shadow mask $\mathbf{M}_S$ as follows:

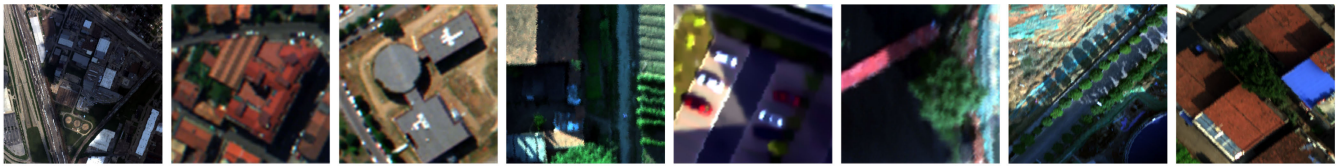
$$\mathbf{M}_{\mathrm{NS}} = (\mathbf{M}_S \oplus \mathbf{O}_{\mathrm{SE}}) - \mathbf{M}_S \tag{7}$$

$$\boldsymbol{q}_j^n = \sum_{i=1}^{N_2} (\mathbf{I}_j \cdot \mathbf{M}_{\mathrm{NS}}) \Big/ N_2 \tag{8}$$

where $\oplus$ is the morphological dilation operation, and $\mathbf{O}_{\mathrm{SE}}$ is a structural element. Similarly with (6), $\boldsymbol{q}_j^n$ represents the averaged reflectance value outside the shadows. Then, the reflectance difference ratio along the spectral dimension between shadow and nonshadow regions can be expressed by

$$\boldsymbol{r} = \boldsymbol{q}^n / \boldsymbol{q}^s. \tag{9}$$

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 3. Real RGB images of the experimental data. (a) SD1, 340 × 340 × 144. (b) SD2, 80 × 80 × 102. (c) SD3, 85 × 85 × 103. (d) SD4, 200 × 200 × 270. (e) SD5, 120 × 120 × 270. (f) SD6, 100 × 100 × 270. (g) SD7, 310 × 310 × 270. (h) SD8, 240 × 240 × 270.

Based on the reflectance difference ratio $\boldsymbol{r}$ with a size of $D \times 1$, the exposure parameter $\boldsymbol{\varepsilon}$ can be calculated as follows:

$$\boldsymbol{\varepsilon} = \boldsymbol{r}\alpha. \tag{10}$$

Coefficient α is produced according to the illumination difference between nonshadow and shadow regions in the brightness image. Specifically, the real RGB images of the input HSI $\mathbf{I}$ and the overexposure image $(\boldsymbol{r} \cdot \mathbf{I})$ are changed into HSV color space. The brightness images corresponding to the value channel of $\mathbf{I}$ and $(\boldsymbol{r} \cdot \mathbf{I})$ are denoted as $\mathbf{B}$ and $\mathbf{B}^E$, respectively. Then, α can be expressed as the local difference between $\mathbf{B}$ and $\mathbf{B}^E$

$$\begin{cases} v^n = \sum_{i=1}^{N_1} (\mathbf{B} \cdot \mathbf{M}_{\mathrm{NS}}) \Big/ N_1 \\ \alpha = v^n \Big/ v^s, \quad v^s = \sum_{i=1}^{N_2} \left(\mathbf{B}^E \cdot \mathbf{M}_S\right) \Big/ N_2 \end{cases} \tag{11}$$

where v^n is the averaged brightness value of the nonshadow region around shadows in $\mathbf{B}$, and v^s is the averaged brightness value of the shadows.

After the above exposure parameters calculation process, the overexposure image is generated as $\mathbf{F} \cdot \hat{\boldsymbol{\varepsilon}}$. The degree of exposure $\hat{\boldsymbol{\varepsilon}}$ is the band averaging result of $\boldsymbol{\varepsilon}$ as (5) calculated, which has the same dimension with $\mathbf{F}$. Moreover, it is worthy of note that the feature extraction is independent of illumination conditions. Thus, the original feature image $\mathbf{F}$ is enhanced with the overexposure process directly, which also can reduce the computation cost and noise interference. Finally, the SGCEM is performed on the simulated overexposure image to identify the targets contained in shadows. For convenience, the detection result of the overexposure image is denoted as $\mathbf{P}_S$.

B. Decision Fusion

The preliminary detection results of the original feature image $\mathbf{P}_N$, and the overexposure image $\mathbf{P}_S$ are obtained with the above detection process. In this step, the preliminary detection results are fused to generate the final detection result.

Since the pixel values of nonshadow regions in the overexposure image deviate from normal values, the nonshadow region result of $\mathbf{P}_S$ is not accurate. Besides, many target pixels in the shadow region cannot be identified in $\mathbf{P}_N$. Considering these issues, the final result is designed as a combination of $\mathbf{P}_N$ and the shadow region of $\mathbf{P}_S$. Specifically, the fusion rule is declared as follows:

$$\mathbf{P} = \mathbf{P}_N + \mathbf{P}_S \cdot \mathbf{M}_S \tag{12}$$

where $\mathbf{M}_S$ is the shadow mask. Moreover, a mean filter with a 3 × 3 window size is further performed on $\mathbf{P}$ to remove isolated pixels and make the result smoother.

IV. Experiments

This section introduces the experimental setting first. Then, the effectiveness of the overexposure image and SGCEM is analyzed. Besides, eight HSIs containing targets both in shadows and well-illuminated regions are utilized to verify the performance of different detection methods. Finally, the detection performance of different methods is compared and analyzed.

A. Experimental Setting

1) Experimental Data: As Fig. 3 presented, eight real-world airborne HSIs with different sizes are utilized as the experimental data. It can be seen that these HSIs are affected by different levels of shadow. To achieve better visual effects, the images in Fig. 3 are enhanced with linear stretching transformation.

SD1–SD4 are cropped from four public data. Specifically, SD1 with the size 340 × 340 × 144 is cropped from the Houston dataset published by the 2013 Data Fusion Contest of the IEEE Geoscience and Remote Sensing Society. SD2 and SD3 are obtained from Pavia Center and University data. The spectral dimensions are consistent with the original data, the spatial sizes of SD2 and SD3 are 80 × 80 and 85 × 85. SD4 is cropped from WHU_Hi_Hanchuan data [37], which contains 270 bands of size 200 × 200. The first four bands are discarded due to the noise. More information about these data can refer to the corresponding paper and links.

SD5–SD8 are acquired at Hunan Institute of Science and Technology, Yue Yang, China, on December 3, 2019, by the Headwall Nano-Hyperspec sensor equipped with the UAV platform. Besides, the flight height is 400 m above ground level. Thus, the ground sampling distance is about 0.24 m. SD5–SD8 comprise 270 spectral bands ranging from 400 to 1000 nm with different spatial sizes. Necessary preprocessing, such as radiometric calibration, has been finished by HyperSpec software with auxiliary information.

The ground truth maps of the above data are mainly obtained by visual interpretation, except SD4 which is well labeled in the raw public data. Besides, the prior target spectrum of each data is calculated as the averaged spectrum of pixels with a fixed 3 × 3 window size in the well-illumination region. The specific locations of these windows with pure pixels are also labeled in the ground truth map of each data with red color for better comprehension.

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2) Quantitative Evaluation: For validation and comparison, the receiver operating characteristic (ROC) curve, the area under the curve (AUC) value, and the separability map have been calculated and drawn. The ROC curve offers a threshold-free performance comparison. In the coordinate system of the ROC curve, false-positive rate (FPR) and true-positive rate (TPR) are designed as the horizontal axis and the vertical axis, respectively. The values of FPR and TPR vary with the threshold in the detection result. The curve closer to the top left corner indicates the result is better. The AUC value calculates the area enclosed by the ROC curve, which can provide a quantitative evaluation of the curve. AUC value is in the range of 0–1. The larger the value, the better the detection result.

The separability map presents the separability of target and background pixels, which is expressed by the Box and Whisker plot. Specifically, the statistical property of pixel values in the normalized detection result is presented by the Box and Whisker plot. Each box covers the value range from upper quartile to lower quartile. In this work, the pink boxes represent the range of the target pixels, while the blue boxes represent the background pixels. Besides, the line in the boxes expresses the median pixel value of different objects. The distance of the two boxes in the y-axis direction reflects the difference in pixel values between the target and the background, which can be used to analyze the separability. A larger distance between two boxes in the y-axis direction indicates the target can be separated from the background more effectively.

3) Compared Methods: In order to fully evaluate the performance of the proposed method, six representative HTD algorithms and two DL-based hyperspectral classification algorithms are utilized for comparison. The HTD algorithms include SAM [13], ACE [16], CEM [20], ECEM [22], hCEM [21], and G2LHTD [6]. Among these algorithms, SAM measures the spectral angle to identify targets. ACE, CEM, and its variants (ECEM, hCEM, and D^{2}CEM) have been introduced in Section I. G2LHTD utilizes an extended morphological attribute profile to extract spatial features which are provided to D^{2}CEM for detection subsequently. The recently published method, G2LHTD, is adopted as a compared method due to its effectiveness on the universal data constructed for HTD.

To compare the performance of different methods more comprehensively, two DL-based hyperspectral classification algorithms are further used for comparison. Specifically, the two state-of-art DL-based methods are robust self-ensembling network (RSEN) [51] and spectral–spatial feature tokenization transformer (SSFTT) [52]. They can capture deep features of different objects and achieve target detection in a different way. Specifically, RSEN [51] aims to use unlabeled data to improve object classification performance. Besides, RSEN consists of a base network and an ensemble network, which can learn from each other with the self-ensembling mechanism. SSFTT [52] uses the transformer model as the backbone and focuses on extracting high-level semantic features for classification. Besides, the self-ensembling and attention mechanisms are widely used in classification and have been proven to be effective [53], [54]. RSEN and SSFTT regard the target detection task as the binary classification task. To train the models, nine target pixels in nonshadow regions and nine background pixels are randomly selected as the training data. Finally, the target probability maps output by RSEN and SSFTT are considered as their detection result.

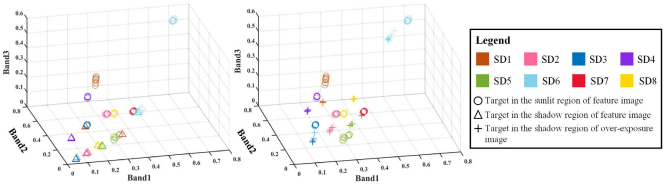


Fig. 4. Effect of the overexposure strategy on target recognizability. For a distinct and intuitive illustration, only the first three bands of SD1–SD8 are presented. Left: difference between the target pixels in the shadow and nonshadow areas of the feature image. Middle: difference between the target pixels in the nonshadow areas of feature image and the shadow areas of overexposure image. Right: legend.

The parameters of the above methods are mainly set according to corresponding papers. Besides, some parameters are also adjusted to ensure that the comparative methods work properly for a fair comparison. For the hCEM, deviation δ is set to 10^{-2}, and λ is set to 400 for SD1, and SD5–SD8. Besides, some key parameters (the regularization parameter λ and the number of layers) of ECEM have been adjusted to achieve a satisfactory performance for each data.

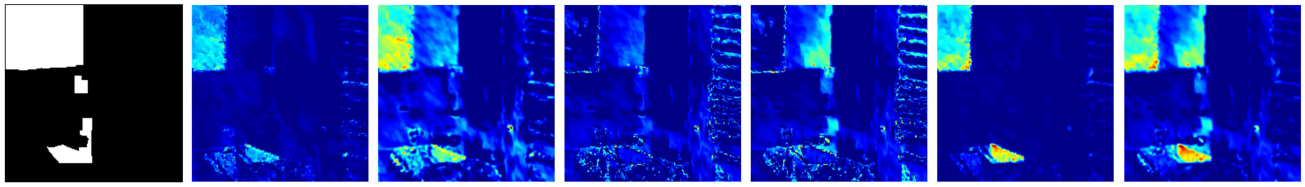
B. Effect of Overexposure, SGCEM, and Decision Fusion

The effectiveness of overexposure, SGCEM, and decision fusion strategy are investigated in this part. Specifically, the distribution maps of the targets in the original image and the overexposure image are compared. Besides, the performance improvement in terms of detection accuracies using SGCEM in the original feature image and the overexposure image are analyzed individually.

For an intuitive illustration, Fig. 4 gives the distribution maps of the targets in SD1–SD8, including the targets in shadow and nonshadow areas, as well as the targets in the shadow area of the corresponding overexposure image. Obviously, the target in shadows (denoted by Δ) is far away from the sunlit target (denoted by $\bigcirc$) in the feature space. Nevertheless, the target feature in the shadow area of the overexposure image (denoted by $+$) is much closer to the sunlit target feature.

To verify the effectiveness of SGCEM and decision fusion strategy, stepwise detection with CEM and SGCEM are performed on SD4, respectively. Corresponding accuracies are reported in Table I. It can be observed that SGCEM is superior to CEM in the detection results of different images and the final detection results, which prove the improvement of CEM in this work is effective. In addition, compared to the detection results of original feature image and overexposure feature image, the decision fusion result achieves the highest accuracy. Therefore, the decision fusion strategy is also useful for the proposed method. Moreover, Fig. 5 displays the corresponding detection result maps of SGCEM and CEM on SD4, which further validates the effectiveness of SGCEM and the proposed detection framework. As shown in Fig. 5, SGCEM finds more

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(a) (b) (c) (d) (e) (f) (g)

Fig. 5. Stepwise detection results of CEM and SGCEM on SD4. (b) and (c) use the feature image for detection, and (d) and (e) use the overexposure image for detection. (a) Ground truth map. (b) CEM result of the sunlit region. (c) SGCEM result of the sunlit region. (d) CEM result of the shadow region. (e) SGCEM result of the shadow region. (f) Fusion result of CEM. (g) Fusion result of SGCEM.

TABLE I
AUC VALUES OF STEPWISE DETECTION WITH CEM AND SGCEM ON SD4. THE BOLD ENTITIES DENOTE THE BEST ONE

Detector	Result of the feature image	Result of the over-exposure image	Decision fusion result
CEM	0.7359	0.7117	0.9429
SGCEM	**0.8027**	**0.7363**	**0.9792**

targets with higher confidence than CEM when the input image is the same. Using the overexposure feature image, the targets hidden in shadows can be found [see Fig. 5(d) and (e)]. To sum up, the proposed overexposure image generation method is helpful to alleviate the spectral variation problem caused by shadows. The SGCEM and decision fusion strategy can improve the detection accuracy.

C. Detection Results of Different Methods

The normalized detection maps of all compared methods are presented in Fig. 6. Besides, the quantitative results of various methods in terms of AUC values, ROC curves, and separability maps are given in Table II and Figs. 7 and 8, respectively.

For these compared methods, the AUC of SAM is relatively higher on SD4–SD8, which indicates the targets in the nonshadow region are homogeneous. Obviously, SAM cannot find the targets in shadows [see Fig. 6(c)]. According to the SAM result of SD1, it can be seen that the spectral heterogeneity of the white roof is more seriously. Due to the spectral heterogeneity, most detection methods do not perform well on SD1. However, with the ability of extracting deep features and more training data, RSEN and SSFTT achieve better performance.

In general, ACE and G2LHTD have lower and similar AUC means, indicating that their overall performance is similar. One possible reason for the poor performance of ACE is that the background distribution is changed when shadows exist. The background distribution is not consistent with assumptions. In contrast, CEM performs better than other traditional detectors, especially on SD5. Although there are noisy pixels in the result of CEM, the targets have relatively higher confidence in its results, which can be seen in the pink boxes of Fig. 8. ECEM and hCEM achieve a better background suppression performance than the original CEM, which can be observed from Fig. 6. Overall, the performance of ECEM is more stable and better than most compared methods. Among these traditional detectors, ECEM obtains the optimal AUC value on the experimental data, except SD5. According to Fig. 6(h), some background pixels have higher values in the result of ECEM on SD5. As for hCEM, due to the interference of slight noise in the image and insufficient utilization of spatial information, some target pixels are also suppressed during the background suppression process, the accuracy of hCEM is not always better than that of CEM. Besides, the detection maps of G2LHTD are smoother than other compared methods, since it extracts morphological attribute profiles for detection. However, G2LHTD lost many targets and does not perform well on HSIs with shadow, since G2LHTD focuses on using global and local spatial information to improve detection accuracy, which cannot solve the spectral variation problem.

The two DL-based methods RSEN and SSFTT can achieve better detection performance than other studied methods in terms of AUC value and ROC curves. Specifically, RSEN and SSFTT obtain the highest detection accuracy on SD1 and SD2, respectively. By comparing the target detection maps and the separability maps of all methods, it also can be observed that the target pixels have higher pixel values in the detection maps of RSEN and SSFTT. Besides, the detection maps of RSEN [see Fig. 6(i)] are different from other methods, which is related to the working mode of RSEN. However, the high-level semantic features extracted by SSFTT are more effective in enhancing the difference between the target and background. Although RSEN and SSFTT give higher confidence for target pixels, they still lose many targets in shadow regions, such as the red car in SD5.

Based on Table II, all compared methods are shadow sensitive and have lower AUC generally, while the proposed method obtains the optimal values in terms of average AUC and accuracies in most cases. In the last row of Table II, the average AUC of the proposed method is 0.9513, while the average AUC of the suboptimal method is only 0.8921. According to the visual inspection of Fig. 6, the proposed method finds target pixels in shadows successfully. Compared with other HTD methods which only can identify the target in nonshadow regions, the target of interest is more obvious and easily distinguishable from the background in the proposed method as well.

The ROC curves comply with the above observation of AUC. According to Fig. 7, it can be observed that the ROC curves of the proposed method are closer to the left top of the coordinate system in all data. Furthermore, the curves of the proposed method and ECEM have a similar location on the results of SD1–SD8, which indicates that the proposed method and ECEM obtain more stable performance.

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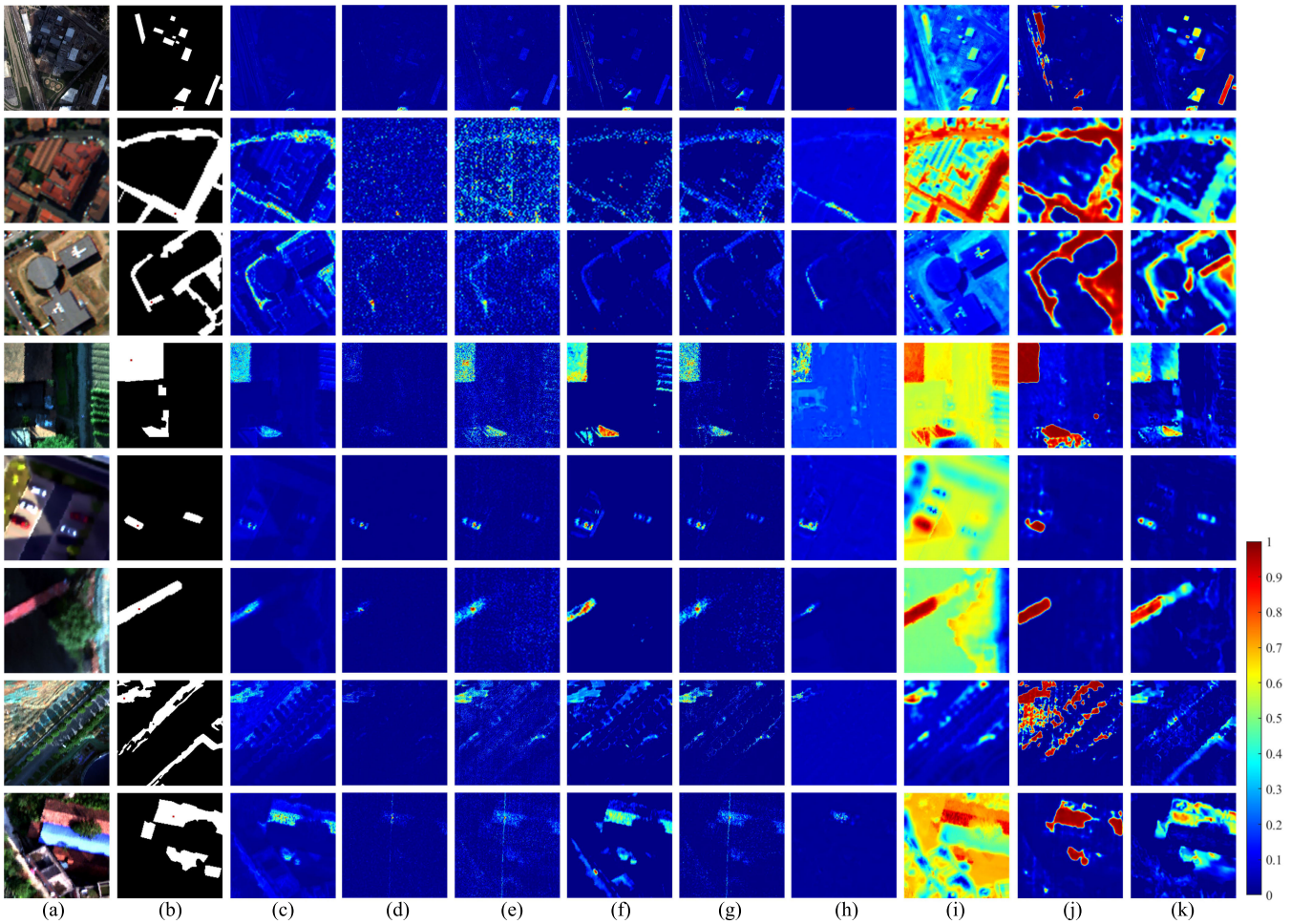


Fig. 6. Detection results of different methods, the red boxes in (b) indicate the pixels used to obtain the prior target spectrum. (a) SD1–SD8. (b) Ground truth map. (c) SAM. (d) ACE. (e) CEM. (f) ECEM. (g) hCEM. (h) G2LHTD. (i) RSEN. (j) SSFTT. (k) Ours.

TABLE II
AUC VALUES OF DIFFERENT METHODS ON EIGHT HSIS WHICH CONTAIN SHADOWS. THE BOLD ENTITIES DENOTE THE BEST ONE

Data	SAM [13]	ACE [16]	CEM [20]	ECEM [22]	hCEM [21]	G2LHTD [6]	RSEN [51]	SSFTT [52]	Ours
SD1	0.7298	0.7085	0.7945	0.8733	0.7316	0.5963	**0.9632**	0.8439	0.9317
SD2	0.5619	0.5178	0.6028	0.8292	0.8029	0.6692	0.8967	**0.9164**	0.9021
SD3	0.7358	0.5786	0.6584	0.8754	0.7624	0.7684	0.7396	0.8976	**0.9176**
SD4	0.8459	0.6384	0.7141	0.8347	0.7166	0.6500	0.8563	0.9143	**0.9792**
SD5	0.8778	0.8991	0.9214	0.8732	0.8613	0.6513	0.9415	0.8305	**0.9981**
SD6	0.8533	0.7179	0.7913	0.8537	0.6860	0.5924	0.9623	0.8596	**0.9741**
SD7	0.7238	0.6126	0.7458	0.8758	0.6498	0.6206	0.8060	0.6863	**0.9211**
SD8	0.8645	0.5955	0.7175	0.9112	0.6444	0.6210	0.9715	0.9363	**0.9864**
Mean value	0.7741	0.6586	0.7432	0.8658	0.7319	0.6461	0.8921	0.8606	**0.9513**

For better analyzing the separability, the abnormal points are removed in the separability maps displayed in Fig. 8. Based on Fig. 8, SAM has a limited distance between the target and background with poor separability in general, since SAM just employs a simple measurement for detection. Combined with Fig. 6, it can be noticed that ACE identifies a small part of targets. Many target pixels are mixed in background with similar probability values. Thus, ACE yields poor separability. CEM can highlight targets in most cases, but CEM fails to control the statistical ranges of background pixels. Furthermore, ECEM and hCEM obtain decent separability. The target ranges of the ECEM and hCEM are relatively wide in all data, their background statistical ranges are relatively small as well. Overall, the background statistical range of G2LHTD is small, but the target value is also suppressed at a lower level, especially in SD6–SD8. RSEN and SSFTT have higher probability values of targets than other methods, but cannot maintain good separation between the target and background in all scenes. In contrast, the gap between the pink and blue boxes is always larger in our method, which means the target and the background are better separated. The strong target-background separability ability of the proposed method

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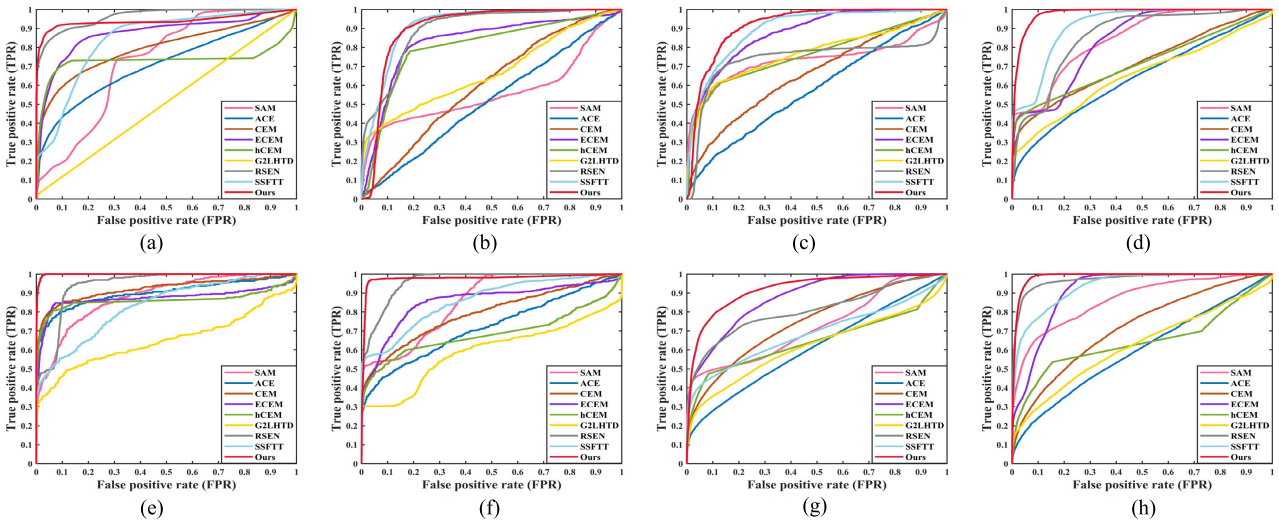


Fig. 7. Detection performance comparison in terms of ROC curves. (a) SD1. (b) SD2. (c) SD3. (d) SD4. (e) SD5. (f) SD6. (g) SD7. (h) SD8.

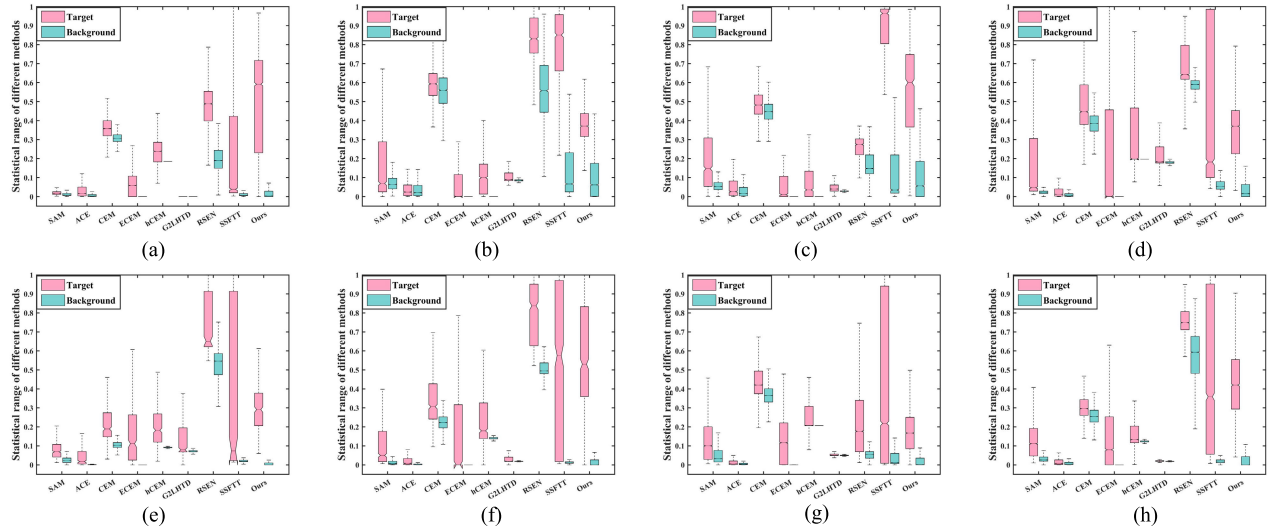


Fig. 8. Detection performance comparison in terms of separability maps. (a) SD1. (b) SD2. (c) SD3. (d) SD4. (e) SD5. (f) SD6. (g) SD7. (h) SD8.

TABLE III

TIME CONSUMPTION OF DIFFERENT METHODS (IN SECONDS). THE BOLD ENTITIES DENOTE THE BEST ONE

Data	SAM [13]	ACE [16]	CEM [20]	ECEM [22]	hCEM [21]	G2LHTD [6]	RSEN [51]	SSFTT [52]	Ours
SD2	0.0473	0.0172	**0.0016**	1.3079	0.0152	46.2982	25.1347	1.1881	0.1678
SD4	0.2851	0.4528	**0.0561**	15.8886	0.2191	292.5180	41.3329	1.2332	0.6356

is owing to the SGCEM which improves the performance of CEM and gives higher probability values of targets. The decision fusion procedure further highlights targets as well as suppresses backgrounds simultaneously.

According to the above analysis, it can be concluded that the traditional detectors have lower robustness to spectral variation, particularly in the natural scene where noise and shadow exist. In addition, the proposed method is comparable with all methods both in subjective and objective evaluation. Through this experiment, the shadow-insensitive of the proposed method is confirmed. The separability maps further prove the superior performance of the proposed method.

D. Time Cost Comparison

The computing time of each method on two images of different sizes is reported in Table III. It can be seen that CEM takes the shortest time. ECEM, hCEM, and the proposed method can be considered as a multiple CEM detection process. Thus, these methods need more time than CEM. Besides, G2LHTD requires the longest computing time since it spends a lot of time on local detection with a dual window detection strategy. Generally speaking, the computing cost of the traditional detectors is similar. As for the DL-based methods, RSEN needs more time than SSFTT due to its ensemble learning process. Moreover, the calculation time

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of the proposed method is competitive. Taking SD2 data for example, the proposed method costs around 168 ms, while ECEM and SSFTT require around 1308 and 1186 ms, respectively. Moreover, it should be noted that the experiments are carried out with the 3.7-GHz Intel Core i9 CPU and NVIDIA GeForce GTX 3090 GPU.

V. Conclusion

This work is a meaningful attempt at the study of shadow-insensitive HSI target detection, which utilizes exposure fusion to solve the spectral variation problem caused by shadows. Specifically, the proposed method utilizes band averaging and total variation to realize feature extraction, which can remove the noise and texture interference on the spectrums to a certain extent. Then, an overexposure image is generated to relight shadow regions in the feature image. The original feature image and the overexposure image are separately used for detection with SGCEM to obtain preliminary detection results. Finally, the detection results are fused with a decision fusion strategy. Experiments executed on real-world HSIs prove that the proposed SGCEM can achieve better background suppression performance than CEM. Furthermore, the proposed overexposure image generation method can alleviate the spectral variation problem caused by lower illumination.

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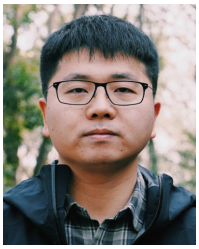
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